

HR Analytics Report

IBM HR Employee Attrition Dataset — Analysis Summary

1. Introduction

HR Analytics (also known as People Analytics or Workforce Analytics) is the practice of using data, statistics, and analytical techniques to understand and improve workforce decisions. Rather than relying purely on intuition, HR Analytics helps organizations answer critical questions such as why employees leave, what drives engagement, and which factors influence performance and retention.

This report is based on the **IBM HR Analytics Employee Attrition dataset**, a widely used dataset for practicing HR data analysis. It contains records of **1,470 employees** across **35 attributes**, covering demographics, compensation, tenure, job satisfaction, performance, and — most importantly — whether each employee left the company (**Attrition**).

The goal of this analysis was to calculate a series of Key Performance Indicators (KPIs) using Python and pandas, in order to identify patterns and factors linked to employee attrition. The findings below are presented in a simple question-and-answer format so each KPI is easy to understand at a glance.

Metric	Value
Total Employees	1,470
Total Attributes (Columns)	35
Employees who Left (Attrition = Yes)	237
Employees who Stayed (Attrition = No)	1,233
Overall Attrition Rate	16.12%

2. Key Performance Indicators (KPIs)

Each KPI below is written as a question an HR leader might ask, followed by a clear, easy-to-understand answer based on the analysis.

KPI 1: Overall Attrition Rate

Q: What percentage of employees have left the company overall?

A: Out of 1,470 employees, 237 have left — giving an overall attrition rate of **16.12%**. In simple terms, roughly 1 out of every 6 employees has left the company. This number becomes the baseline against which every other KPI is compared.

KPI 2: Attrition Rate by Department

Q: Which department loses the most employees?

A: **Sales has the highest attrition rate (~20.6%)**, followed by Human Resources (~19%), while **Research & Development is the most stable (~13.8%)**. This tells us attrition is not evenly spread — Sales is the department facing the biggest retention challenge.

KPI 3: Attrition Rate by OverTime

Q: Does working overtime affect whether an employee leaves?

A: Yes, significantly. Employees who work overtime leave at roughly **2 to 3 times the rate** of employees who don't. This pattern holds true in every single department, making overtime one of the strongest predictors of attrition in the entire dataset.

KPI 4: Attrition by Department + OverTime Combined

Q: Where is the overtime effect the strongest?

A: **Sales employees who work overtime have the highest attrition rate of all — around 37.5%**. That means more than 1 in every 3 overtime Sales employees eventually leaves the company, making this the single riskiest combination in the dataset.

KPI 5: Attrition Rate by Job Role

Q: Which specific job roles see the most turnover?

A: **Sales Representative has by far the highest attrition rate (~39.8%)**, followed by Laboratory Technician (~23.9%) and Human Resources (~23.1%). Senior roles like Manager (~4.9%) and Research Director (~2.5%) are the most stable. This shows attrition is concentrated in junior, front-line positions rather than leadership roles.

KPI 6: Average Monthly Income — Leavers vs. Stayers

Q: Do employees who leave earn less than employees who stay?

A: Yes. Employees who left earned an average of **■4,787** per month, compared to **■6,833** for employees who stayed — roughly a **30% pay gap**. This suggests compensation is a meaningful factor in why employees choose to leave.

KPI 7: Average Age — Leavers vs. Stayers

Q: Are younger employees more likely to leave?

A: Yes. Employees who left were, on average, **33-34 years old**, compared to **37-38 years old** for employees who stayed. This indicates that attrition is more common among younger, early-career employees than older, more established staff.

KPI 8: Average Tenure (Years at Company) — Leavers vs. Stayers

Q: Do employees leave early on, or after many years with the company?

A: Employees who left had worked at the company for about **5 years** on average, compared to **7-8 years** for employees who stayed. This means attrition tends to happen earlier in an employee's journey, rather than after they've become long-term, established staff.

3. Overall Story & Conclusion

Bringing all the KPIs together, a clear pattern emerges: employees most likely to leave the company tend to be **young, early in their tenure, work in the Sales Representative role, work overtime frequently, and earn below-average pay**. Overtime stands out as the single strongest factor — its effect on attrition holds true across every department, even in otherwise stable teams like Research & Development.

Recommendations for HR Leadership:

- Reduce excessive overtime, particularly within the Sales department, as this shows the strongest link to attrition.
- Review compensation for junior and entry-level roles, especially Sales Representatives, since leavers earn notably less on average.
- Focus retention efforts on the first 3-5 years of employment, since attrition risk is concentrated in early tenure rather than among long-serving employees.